

identification news

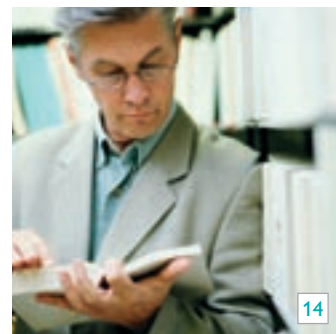
on the move



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PHILIPS

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Philips scores in German stadiums

The 2006 FIFA World Cup™ will be the sporting event of this year. The best players from the top footballing nations will gather in Germany to display their skills. And off the field, another star is playing an important role in the competition – Philips MIFARE Ultralight IC! Fitted into all the match tickets for this year's FIFA World Cup™, the contactless smart card chip will ensure secure and fast access to each of the 12 stadiums hosting the games.



Four stadiums holding FIFA World Cup™ football matches use Philips' MIFARE contactless infrastructure – Cologne, Munich, Frankfurt and Hannover. These proving grounds have shown how robust and reliable the solution is, with the system used during other major events such as the FIFA Confederations Cup 2005 and soccer championship games.

Contactless infrastructures are also being installed at the remaining FIFA World Cup™ stadiums to enable them to read the MIFARE Ultralight-equipped tickets. They will join the many sport and leisure facilities around the globe that enjoy the benefits of our contactless smart card technology: benefits including easier and safer access control, plus support for additional services such as loyalty rewards, e-purse capability and other applications.

As an official sponsor, Philips will be playing its part in making this the most enjoyable FIFA World Cup™ yet. Whether you are lucky enough to be in the stadium or enjoying the game with a few friends at home, our technical expertise and advanced products will make it an experience to remember!

A low cost solution that's easy to integrate, MIFARE Ultralight allows high volume integration of smart card tickets into existing ISO 14443A-compliant systems, making it the ideal replacement for event ticketing infrastructures using traditional paper or magnetic stripe tickets. Low cost smart paper tickets offer manufacturers competitive advantages, including access to established high volume markets through the existing contactless MIFARE base. It's versatile too – as well as event ticketing, MIFARE Ultralight is used in transport systems all over the world.

For more information on:

MIFARE products: www.philips.semiconductors.com/products/identification/mifare

Philips and the 2006 FIFA World Cup Germany™: www.philips.com/fifaworldcup

NFC: come closer, go further

Wouldn't it be great to instantly swap contact details with a new friend or business associate?
How about watching a clip from the film on the poster that caught your eye? Or having all your travel details and tickets stored on one device that also lets you check in for your plane and hotel?
Wouldn't it be even better if we could do all this anywhere and at anytime, with no fuss and no bother?



Well, now you can. What's more, you can do it in the simplest of all ways – with a touch.

Welcome to the world of Near Field Communication (NFC). A world where queuing is a thing of the past, travel is stress free and you can easily share precious memories with loved ones. NFC helps boost productivity at work and lets you have fun downloading anywhere.

Touch, interact, share

NFC lets you connect and share information between electronic devices simply by bringing them close together. >

NFC technology

NFC evolved from a combination of contactless identification and connectivity technologies. It offers an easy, intuitive way to connect devices and share data between them. Operating over short distances and not requiring any configuration by the user, NFC is the ideal way for consumers to interact with the connected world around them via their cell phones.

Standardized under ECMA-340 (ISO/IEC 18092), NFC operates at 13.56 MHz. It's fully compatible with the Philips MIFARE and Sony FeliCa contactless smart card platforms. So it can be used immediately with existing infrastructure, speeding uptake and avoiding the 'chicken-and-egg' problem that often hampers new technologies.



It's as simple as that. Just touch and interact. For example tap two NFC-enabled phones together to start a conference call or swap phone numbers instantly. When you're shopping, put your phone against an interesting DVD to see a short trailer. Or see a poster for a concert and straightaway you have a reminder in your phone. How? Just by touching the poster with your phone.

With NFC, connection and interaction is fast, easy and intuitive. And it's secure too (see box 'Secure transactions'). So you can confidently use your phone as an electronic wallet or payment card. You won't just

use your phone to remind you about that concert, you'll also use it to buy the tickets and share them out amongst your friends.

Easily built into cell phones, PDAs and other handheld devices, NFC opens the door to an almost limitless world of new services and business opportunities. It has the potential to make all areas of our work and home lives easier. We are only just beginning to consider where NFC can take us. >

Secure transactions

Before people use any new method of payment, they want to be sure it is secure. You can easily guarantee secure transactions over NFC by combining the NFC chip with a SIM card or secure smart card controller like Philips' SmartMX. This allows you to offer cell phones with features such as 3-DES encryption, EMV (Europay-MasterCard-Visa) compliance and support for other secure transaction environments.

All of our NFC modules are designed to be easily built into this kind of arrangement. And to make it even easier for you to develop secure NFC applications, we've created a reliable, high-security interface called S²C. Providing the link between your 'security IC' and NFC module, S²C is fully compliant with all modes, types and data rates of ECMA-340 (ISO/IEC 18092) and ISO/IEC 14443.

Meeting people's needs with NFC

NFC has the power to make people's lives easier. Together with other industry leaders from across the value chain, Philips is working hard to make sure NFC is implemented in a way that maximizes the effect. As an industry, we need to make sure NFC devices are easy to use and offer the services consumers want.

To this end, Philips helped establish the NFC Forum. This non-profit organization aims to promote the standardization of NFC technology and the interoperability of NFC-enabled devices. Both these factors are vital to promote widespread uptake of a new technology.

What consumers want

Of course, to deliver what consumers want and need, we need to know what that is. So we've joined forces with VISA to carry out usability studies. These tests looked at consumers' opinions of the NFC concept and technology and how they would like to see it presented. The results (see box 'Usability studies') will form the basis for guidelines on how best to implement NFC to meet consumers' needs and expectations.

But lab-based studies are only one part of the story. We're also involved with various partners in a number of field

tests. These trials cover a number of different applications and are taking place all around the world.

For example, Hanau (Germany) and Taipei are both hosting public transport ticketing pilot schemes. Sports fans in Atlanta are using NFC within the stadium to make purchases from concession stands and access mobile content such as ringtones and wallpapers. Similarly, season ticket holders at the soccer club Roda JC in the Netherlands are using NFC-enabled phones to get into the stadium and buy refreshments and merchandise.

In Northern France, 200 residents of Caen are taking part in the most-widespread NFC trial to date. Via their NFC-enabled Samsung D500 phones, these trailblazers can shop in local supermarkets and stores, pay for car parking, use banking applications or access information on local sites. Launched on October 18 2005, the trial is a joint project between Philips, France Telecom, Orange, Samsung, Groupe LaSer and Vinci Park.



Field tests such as these will help us understand how NFC stands up to the rigors of the real world. They also show us how and where NFC can most benefit people in their everyday lives. Armed with this knowledge, we can develop NFC-enabled services and devices that meet consumers' expectations and, more importantly, make all our lives easier. >

Usability studies

In December 2005, Philips and Visa invited a number of volunteers to a usability test facility in Atlanta, USA. They were there to examine consumers' expectations of and reactions to NFC applications, in particular contactless payment.

The tests showed:

- NFC is easy to learn and to use
- It is convenient and saves time
- People enjoy downloading content from smart posters
- A consistent logo should be used everywhere NFC is available and should indicate where to place the NFC-enabled device
- The 'touch' should be the first step in any interaction to receive information
- People like paying with NFC-enabled phones but need to feel they are secure
- Confirmation is only required for payment and private data applications



NFC on the bus

The southern German city of Hanau near Frankfurt was the site of one of the first large-scale NFC trials. Started on May 2, 2005, the trial was set up by a group of partners including Philips, Nokia and the local transport authority the Rhein-Main-Verkehrsverbund (RMV). It allowed 200 regular local bus passengers buy, store and use tickets via a Nokia 3220 cell phone equipped with a special NFC shell.

Preliminary results from a market research study on the trial are already available and are very encouraging. NFC mobile ticketing proved to be extremely popular, with users describing it as easy-to-use, convenient, fast and trouble-free. According to some users, friends of all ages were keen to get hold of the handsets. Following the success of NFC on the buses, the partners are looking at extending the trial to include further applications.

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China: a key market for NFC

Easy-to-use, versatile and compatible with existing contactless ticketing infrastructures, NFC is clearly a technology with the potential to revolutionize people's lives. And in greater China, NFC has a market with equal potential.

A unique combination of factors makes China such an attractive market for NFC. Firstly, the country already has extensive MIFARE infrastructure that NFC-enabled devices could use. Some 80 cities have MIFARE-based transport ticketing programs with over 200 million cards deployed. By contrast, the credit card system is not as

developed as in some other markets, leaving the way clear to move straight to contactless transactions.

In addition, the cellular and financial landscapes in mainland China are still quite regulated with just two major network operators and one large central bank. This makes

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NFC at 3GSM

Attendees at this year's 3GSM exhibition in Barcelona, Spain had the chance to try out NFC for themselves. Hostesses greeted visitors, registering and authenticating their identity badges via NFC-enabled PDAs. Following the display on their PDA, the hostess invited 200 randomly pre-selected visitors to pick up a free NFC-enabled phone from the Philips booth. The lucky 200

could then use the phone to read their own badge, exchange business cards, enter a Philips sponsored competition and even buy DVDs at the kiosk.

We'll be running similar demonstrations at many other major trade shows over the course of the year, giving you the chance to experience the joy of NFC yourself.



Taiwan NFC ticketing trial

In October 2005, Taiwanese commuters began trialing NFC on the Taipei public transport network. 100 volunteers, chosen by lucky dip after the scheme attracted huge public interest, are using prototype cell phones from BenQ to buy, store and use tickets. The phones feature Philips NFC and SmartMX ICs.

Building on existing MIFARE infrastructure, this trial is the first step in the Taiwan Proximity Mobile Transaction Service Alliance's (PMTSA) program to

develop NFC applications such as payment, micro-payment and mobile content services. The Alliance includes Acer, BenQ, Chunghwa Telecom, Far EastTone Telecommunication, Taipei Smart Card Co., Taiwan Cellular Corp., Asia Pacific Broadband Wireless Communications Inc., VIBO Telecom Inc., MasterCard International, Visa International and Philips. Uniquely, the government is heavily involved in the roll-out of NFC in Taiwan, pushing for the creation of the PMTSA and driving discussions.

standardization and negotiations easier for all involved. Finally, China has a lot of handset makers and design houses, and a large manufacturing base. So all the links in the value chain are available locally.

Philips is involved in discussion with all the major players. Together we are working on a number of projects to bring the benefits of NFC to the Chinese public. In October 2005, an NFC pilot scheme was launched in Taiwan and trials in mainland China are due to start in the second half of this year. Initially, these trials will concentrate on transport ticketing. Payment and other applications will be added later.

With strong interest from local network operators, financial institutions and handset makers, China is set to become a key market for NFC. The country currently has some 400 million mobile phone users. If China's rapid economic growth continues, could we soon be talking about 400 million NFC users?



Read more about NFC and Philips' NFC products at www.semiconductors.philips.com/nfc



Brazil: balls, bikinis and buses

What comes to mind when you think about Brazil? Football? Carnival? Samba? The Copacabana? Did you know that Brazil is also a booming market for RFID? Philips does. We're involved in more than 60 Brazilian transport ticketing projects. Together, the projects are deploying a total of 10 – 12 million card ICs this year alone.

It shouldn't come as any surprise that the market for RFID in Brazilian transport ticketing solutions is so healthy. Brazil is the 5th largest country in the world in terms of both area and population. It also has the 4th longest road network of any country in the world. And it is the home of São Paulo – the world's 3rd largest city.

Mass transit

São Paulo is the site of our largest transport ticketing project in Brazil. Set up by system integrators Prodata and Digicon, the São Paulo scheme will allow people to travel throughout the city's bus, underground and train network with a single MIFARE-based card – the *Bilhete Unico*. The >



scheme is already fully rolled out on the buses and is also in operation in a few metro stations. It is being extended to include the rest of the underground and train systems this year with an estimated 7.5 million cards in the field.

Prodata is also the system integrator of another large-scale transport ticketing program, this time in Rio de Janeiro. The MIFARE-based RioCard is already accepted on all the city's buses and the scheme is currently being expanded to include the local ferry network. Once that's complete, the system is expected to be extended to include the Rio's minibuses and vans which supplement the bus network. The Rio transport authority has already deployed 2.5 million cards and expects to issue a further million this year.

Our ICs are also being used in projects in other Brazilian cities. Fortaleza will roll out 700 thousand cards, Manaus 600 thousand and Cuiabá 200 thousand. There are also numerous lower-volume schemes (30 – 100 thousand cards) in smaller satellite cities as well as a few intercity bus ticketing projects.

All of these schemes are based on Philips MIFARE technology. Typically, passenger cards feature a MIFARE 1K chip while drivers and conductors carry a card with a MIFARE 4K chip. Sonsun, one of the biggest card manufacturers in Brazil, is a key supporter of MIFARE for transport applications. Ryozo Hamano, President Sonsun stated, "In addition to contactless smart cards based on MIFARE 1K, 4K and Ultralight chips, we also plan to introduce Philips MIFARE DESFire and SmartMX cards for other market segments beyond public transportation. We have no doubts that MIFARE is the best technology for contactless smart cards and that Philips is the leading company in contactless chip cards worldwide."

Brazil facts and figures

- The 5th largest country in the world:
8.5 million square kilometers
- The 5th most populous country in the world:
185 million inhabitants
- The world's 9th largest economy
- Has the 4th largest road network in the world

On the infrastructure side, the card readers are based on the MIFARE contactless reader IC family, particularly the MF RC531. The transport schemes across Brazil are split between three systems integrators: Prodata, Empresa 1 and Digicon. Between them, the three companies installed around 55 thousand readers last year.

More than just bus tickets

Brazil is also looking to exploit ID technology in a number of areas besides transport ticketing. These include electronic passports, access control for government sites, livestock tracking and smart labels in supply chain management. One of the more ambitious projects under way is a student ID program. Eventually, all school children across Brazil will have a smart ID card for use on school buses and to record attendance levels. Being rolled out this year, the scheme will initially deploy 17 million MIFARE-based cards rising to 50 million when completed.

With figures like that, and huge potential for further growth, it's no wonder people in the know are calling Brazil 'the little China'.

For more information on MIFARE:

www.philips.semiconductors.com/products/identification/mifare/

Cards 2006



Come and visit us at Cards 2006 - the 11th international congress and exhibition on cards, services and technologies in Brazil - and see our highly secure MIFARE DESFire smartcard controllers, contactless reader ICs and eGovernment solutions in action for yourself. Brazil's largest smart card event, Cards 2006 runs from April 3 to April 5 at the Frei Caneca Convention Center in São Paulo. You'll find us at booth 17.

“Our customers need to trust us as well as our product. They demand an impressive track record and a commitment to the market.”

Partner perspective

Our third Partner Perspective takes a look at **ACG Identification Technologies**, an independent component and technology supplier active throughout the value chain of smart card and RFID-based systems. CEO Ulrich Reutner and CTO Peter Kronegger tell us more about ACG – and in particular, its ID division.



Ulrich Reutner, CEO

ACG

Established in 1996, ACG provides smart card, transponder and reader manufacturers with chips, modules and pre-laminated inlays; and system integrators with inlays, readers, software and sub-systems. “Essentially, we work with our customers in two ways,” says Ulrich Reutner, “either acting as a component supplier or working together to produce tailored solutions that meet their specific needs. Reflecting this, in 2005 we formed two divisions – ACG Distribution and ACG ID – so we could guarantee our customers the most efficient supply and support.

ACG Distribution

This division distributes a wide range of identification products such as bare silicon chips, smart card modules, pre-laminated inlays and reader ICs for advanced smart cards, transponders and readers. In the distribution world, speed and reliability of delivery are critical. It’s a question of looking ahead – anticipating the markets’ evolving needs to guarantee delivery of the right volumes at the right time.



As an independent supplier, ACG Distribution works closely with several major semiconductor suppliers. “Philips Semiconductors’ portfolio is extremely important to us” outlines Ulrich Reutner. “ACG Distribution significantly increased its turnover with Philips products in 2005, an incredibly good year for contactless smart cards and readers.”

ACG ID

ACG ID was created to meet the growing demand for RFID technology for secure identification applications in a range of markets, focusing on Government applications, Financial Transactions, Public Transport and physical and logical Access Control.

“Our ID division distributes third-party contact readers, and designs its own contactless readers, which are sold as OEM, Plug & Play and desktop versions and as plug-in modules for handheld devices. But the devices themselves are only part of the deal. Our technological know-how (particularly in RFID technology, microcontrollers and reader devices) is matched by our experience in credential-related issues. So we can provide our customers with the ideal combination of ID credentials and reader devices for their applications.”

“Over the years, we’ve developed a complete and in-depth understanding of the application issues and integration challenges related to contactless reader implementation. A long-time partner with both manufacturers and system integrators, we understand the materials, components and manufacturing processes involved, including the technical constraints, quality assurance and reliability issues. So,

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unlike most hardware suppliers, we also offer independent advice on combining components and technologies for specific applications.”

RFID: Convenience and security

The various markets ACG ID serves all have one thing in common: the need for security. “Whether it’s a national border, a financial transaction, a building, an IT network or a public transport system, you need to prevent unauthorized access, without making things difficult for those who do have the right to enter.”

In some of these markets – such as access control and public transport – RFID already has a strong presence. In others, the process is just beginning. With its unique know-how and extensive experience, ACG ID is well-placed to help organizations and countries to make the transition, providing solutions specifically designed to enable a convenient, cost-effective migration to contactless identification.

“The contactless smart card’s unique combination of security and convenience makes it ideal for all kinds of secure personal identification applications, e.g. ePassports, eDriving licenses, financial transactions. Several countries including Belgium, Germany and Sweden have already introduced ePassports and the rest of the European Union is expected to follow by the end of 2006. The next stage will be the introduction of e-ID cards, which Scandinavia, Germany and France are already planning to implement by the end of 2007. And we expect to see other countries extending their eGovernment programs in these and in other areas such as eDriving licenses.”

ACG ID: Reliability

Security isn’t the only key consideration here. For such applications, interoperability and consistent quality are also essential, as Peter Kronegger explains, “The most critical element is the reader. An ePassport reader, for example, has to guarantee high data transfer rates so it can deal with huge amounts of data as well as large numbers of travelers. It also needs to work reliably with a variety of chips and smart card operating systems. With so many card vendors active in the market, governments need to be sure every reader can read every card. If your bus ticket doesn’t work, it can be inconvenient; if your passport doesn’t work, it can be a disaster.”



Making a working sample is difficult in itself, but producing high volumes of reliable readers is even more complex. It takes special expertise, and that’s why ACG’s customers keep coming back. “Our customers need to trust us as well as our product. They demand an impressive track record and a commitment to the market. Of course, we can showcase our long-running partnership with Philips. And our commitment: when we start a project together, we stay with it for its duration. That’s something we’re both happy to do. It’s a very open partnership, built on mutual trust.”

And it’s a personal relationship as well as a professional one. “Philips’ Gratkorn team, for example, is just 20 minutes away from the RFID reader R&D center I head up in Graz,” Peter Kronegger continues. “I’ve known some of them since 1990. We’ve even trained Philips engineers at our SmartWorldAcademy in Prague, where we offer everything that’s required to develop and test a complete smart-card-based solution, including the know-how of our experts.”

“Today, our partnership with Philips is stronger than ever. We’ve succeeded in almost every important project we’ve worked on together, and now we’re stepping up the cooperation in new projects and business development.”

For more information on MIFARE: www.semiconductors.philips.com/products/identification/mifare/index.html



The silent revolution: RFID in libraries

Self-service check-in and check-out. Automatic sorting and cataloguing. Advanced theft prevention. RFID can make life easier for library users and staff alike. NBD|Biblion, a Netherlands-based library supplier, is seeing a growing demand for RFID solutions in public libraries across Europe.

NBD|Biblion's core business is supplying books for Dutch libraries. The books are supplied shelf-ready, complete with plastic cover, spine labels and RFID tags. "Around 100 libraries in the Netherlands have RFID tags on their entire collections and more are switching over to it," explains Maarten Tiebout, RFID project leader at NBD|Biblion. "We're tagging 2.7 million books per year. Altogether we've tagged some 6 – 7 million books, and we've sold a similar number of tags to libraries in the Netherlands, Belgium and France for their existing collections."

The company uses tags based on Philips ICODE SLI ICs. These state-of-the-art yet low-cost RFID solutions can be read at up to 1.5 m and offer read speeds up to 60 labels per second. "We chose Philips' ICODE ICs because, besides numerous other features, they offer the largest reading distance, incorporate open-standard EAS anti-theft protection and guarantee the strict ISO compliance. This

gives libraries much more flexibility in their theft detection systems and flexibility than rival ICs that use proprietary anti-theft solutions," says Maarten.

With RFID, consumers can speed up their visit to the library by checking-out books, CDs and DVDs themselves. They can also serve themselves when making returns, saving library staff time. The returned item is scanned in and automatically sorted by electronic sorting stations.

Due to open in 2007, the new central library in Amsterdam will be the Netherlands' most prestigious library. Thanks to RFID, it will also be one of the most advanced and accessible libraries in Europe.

For more information on ICODE:
www.philips.semiconductors.com/products/identification/icode/

Identification updates

e-ticketing reader for Norwegian State Railways

Norwegian train conductors will soon be checking tickets with pocket PCs fitted with contactless smart card readers as part of a new e-ticketing system. Installed by Arcontia AB, the system is based on leading-edge technology involving the Philips MIFARE DESFire card standard.

The Mobile Terminal (MT) project is an important part of the ongoing KISO project, introducing electronic ticketing for all of Norway's public transport organizations including Norwegian State Railways, Stor-Oslo Lokaltrafikk (buses) and Oslo Sporveier (metro, trams, buses).

Valencia launches contactless smart card ticketing

The body responsible for public transport in Valencia, Spain, the Entitat de Transport Metropolità (eTM) is launching a new contactless smart card ticketing scheme based on Philips MIFARE technology. Following research into similar schemes around the world, the eTM chose MIFARE for its proven reliability and extendibility. The smart card scheme will be used for underground, bus and tram services. It will

be implemented in the central district this year, and extended to other regions by 2007. The scheme will not only reduce congestion and fraud, but also maintenance costs.

For more information on MIFARE:

www.philips.semiconductors.com/markets/identification/products/mifare/

MIFARE DESFire technology powers Madrid's new transport card

The Regional Transport office of Madrid has chosen Philips MIFARE DESFire technology for its new Sube-T transport card providing commuters with faster and safer access to platforms. Initially, MIFARE DESFire chips will be built into annual tickets for the city's underground and stop-train network. A second stage due to start in January 2007 will cover the whole EMT (Madrid local transport) bus network. Between 2008 and 2011, the smart cards will be

gradually implemented across the whole Madrid regional public transport network. Thanks to its large data storage capacity, the card could later be used for other applications such as paying for goods and services or accessing sports and leisure facilities within the city.

For more information on MIFARE DESFire:

www.philips.semiconductors.com/products/identification/mifare/index.html

SMARTMX at heart of New Zealand ePassports

Canadian Bank Note Company Limited, one of the world leaders in identification document issuing systems, has successfully integrated Philips SmartMX contactless smart card chip technology in New Zealand's new electronic passports. Highly secure, SmartMX is certified to the rigorous Common Criteria EAL5+ standard. It will be used to store and protect unique passport information, reducing

fraud and forgery of travel documents and increasing security for travelers. Following similar decisions by the German and Austrian governments, this announcement confirms Philips as the chip supplier of choice for ePassports.

For more information on SMARTMX:

www.philips.semiconductors.com/products/identification/smartmx/

Meet us at the following events

Event: **Cards Brazil**
Date: 3 - 5 April, 2006
Location: São Paulo, Brazil
Website: www.dib.com.br/

Event: **CTIA Wireless**
Date: 5 - 7 April, 2006
Location: Las Vegas, USA
Website: www.ctiawireless.com

Event: **RFID Journal Live!**
Date: 1 - 3 May, 2006
Location: Las Vegas, USA
Website: www.rfidjournal.com/live2006

Event: **RFID World Asia**
Date: 1 - 3 May, 2006
Location: Singapore
Website: www.worldofcards.biz/2006/rfidwa_sg

Event: **Euro ID**
Date: 16 - 18 May, 2006
Location: Cologne, Germany
Website: www.euro-id-messe.de

Event: **Smart Card Conference**
Date: 30 May - 1 June, 2006
Location: Beijing, China
Website: www.smartcards-china.com/

'On the Move' is published by
Philips Semiconductors
Building VS
P.O. Box 218
5600 MD Eindhoven
The Netherlands

Website: www.semiconductors.philips.com/identification

If you would like to receive a regular subscription of On The Move for free, register with www.my.semiconductors.com

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Philips Semiconductors

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Date of release: March 2006
Document order number: 9397 750 15505

Printed in the Netherlands

